

Supplementary Material

Improving HTR Attention Maps with Register Tokens

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This document collects the per-group training curves referenced in Section 5 of the main report and expands the four future-work extensions listed in Section 7.

S1. Per-Group Training Curves

Figure 1 shows the per-group CER convergence for the four experiment blocks not shown in the main report (architecture ablations, pretrained-lane fine-tuning, READ2016 register sweep, synthetic pretraining). The two catastrophically-failing rows of the architecture ablation (no CNN stem, legacy row-major layout) sit as flat lines near 73% from epoch 1 and are visible at the top of the top-left panel. The naive-uniform-LR ViT-B/16 fine-tune stays above 70% throughout the pretrained panel, in contrast to the two-group-LR and LLRD runs that converge cleanly. The READ2016 register variants collapse onto essentially the same trajectory, mirroring the IAM behaviour in the main report. The synthetic-pretraining curves terminate at the cluster walltime rather than at their intended epoch budget; the reported numbers are lower bounds on convergence.

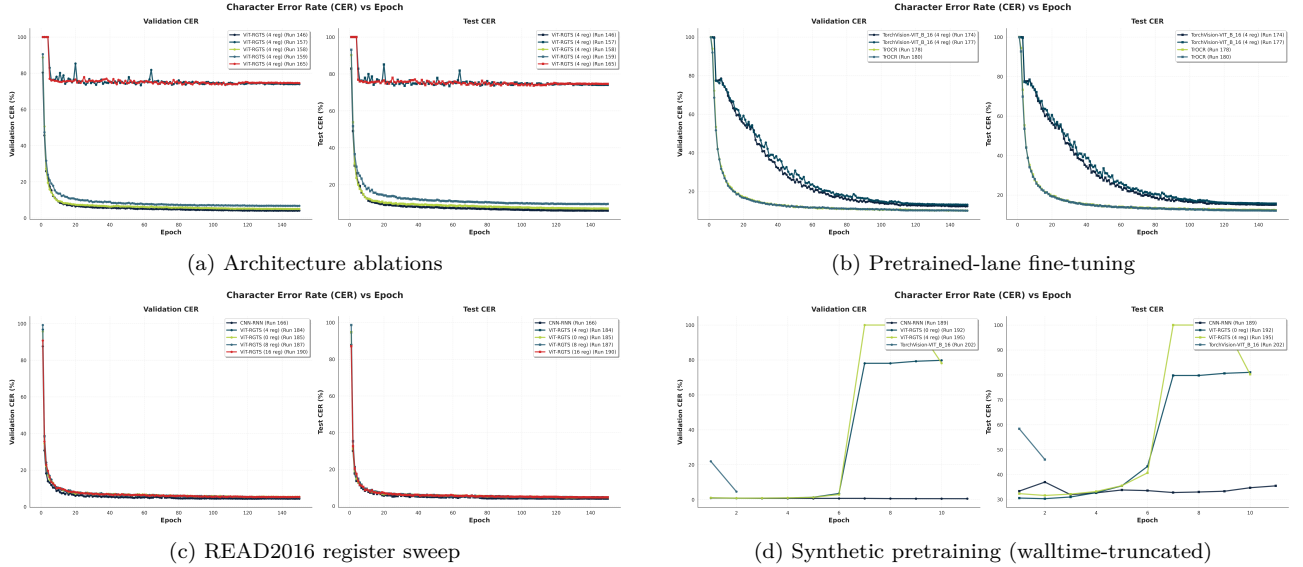


Figure 1: Test-CER convergence for the four experiment blocks not shown in the main report.

S2. Extension Proposals

Writer identification via VLAC over ViT-RGTS attention. The Souibgui *et al.* [2] VLAC construction can be plugged in directly on top of the ViT-RGTS attention output. At inference, for every recognized character in every training-set page, extract the CTC peak column t_c , the patch-token embedding at t_c from the final Transformer layer, and the attention weight vector at the character’s query. Cluster the patch-token embeddings across the training corpus into K character-prototype centroids, assign each character to its nearest prototype, and accumulate residual VLAC-style aggregates per prototype into a per-page descriptor. Evaluate with writer-retrieval mAP, top-1, and top-5 on IAM writer-ID splits. This is the single most direct test of whether the interpretability claim in Section 5.3 of the main report translates into a downstream task.

CLS-augmented CTC ViT. The residual sink pattern documented in Section 5.3 of the main report is attributed to the absence of a CLS-style aggregate query. Adding a single CLS token in front of the register block, supervised at training time by a light auxiliary loss (for example a line-level writer classification head that is unrelated to the CTC objective), should let the registers take over the global-information role more

completely. The predicted signature is a flip of the token-norm asymmetry: registers now carrying more L2 norm than patches, matching the classification-ViT direction Darcet *et al.* [1] report.

Attention-entropy regularizer. A low-weight entropy regularizer on the final-layer patch attention would provide the gradient signal that CTC lacks. Given the set $\mathcal{T}$ of non-blank CTC query positions and Shannon entropy H , the auxiliary loss is $\mathcal{L}_{\text{aux}} = \beta \cdot |\mathcal{T}|^{-1} \sum_{t \in \mathcal{T}} H(\mathbf{a}_t^{(L)})$, added to the primary CTC loss. A starting range for β is $[10^{-3}, 10^{-2}]$; larger values flatten attention rather than sharpen it. The regularizer should be evaluated jointly against CER and against the attention-concentration metrics of Section 4.3, not against CER alone.

Synthetic-to-IAM curriculum. The synthetic-pretraining lane was walltime-truncated. A converged run — pretrain to convergence on the $\sim 950\text{K}$ synthetic corpus, then fine-tune on IAM with a decayed learning rate — would test whether the CER gap between ViT-RGTS and the CNN-RNN baseline closes under pretraining, without altering the register-attention story.

References

- [1] Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers. In *International Conference on Learning Representations (ICLR)*, 2024.
- [2] Mohamed Ali Souibgui, Alicia Fornés, Yousri Kessentini, and Beáta Megyesi. Interpretable writer recognition via vectors of locally aggregated characters. *Pattern Recognition Letters*, 2022.